Job Creation Index vs Job Displacement Charter (JCI-JDC)

Executive summary

The **Job Creation Index vs Job Displacement Charter (JCI-JDC)** is a proposed **measurement-and-governance standard** for organisations and governments to quantify and manage labour-market impacts of advanced automation and AI systems - especially **generative AI** and “high-risk” workplace AI deployments - using **auditable job-flow metrics** paired with a **binding-style charter** that constrains incentives to externalise displacement harms onto workers, communities, suppliers, and states. The framework is designed to be **neutral** (not pro-automation or anti-automation) and instead focuses on **verifiability, comparability, and accountability** for employment outcomes, job quality, and worker transition capacity.

This report operationalises JCI-JDC by combining (i) established labour statistics standards and job-flow measurement practice (notably gross job creation and destruction concepts), (ii) emerging AI governance and disclosure regimes, and (iii) institutional risk thinking consistent with PAD-style dossiers. The report’s structure is adapted from a PAD A-K format and is intended to be updateable and auditable.

Why this is needed now. Modern labour markets exhibit large simultaneous “gross” job flows - jobs are created and destroyed continuously even when net employment changes are small - so net headcount alone is an inadequate indicator of disruption. At the same time, task-based economic research emphasises a core tension: automation and AI can produce a **displacement effect** (capital replacing labour in tasks), but can also create **reinstatement/new-task effects** that expand labour demand (often contingent on complementary innovation and policy choices). Institutional evidence to date is mixed: official analyses and surveys often find limited aggregate employment effects so far, but material risks around job quality, work intensity, privacy, and downstream inequality - particularly as adoption scales.

Core deliverables in this paper.

- A **formal metric suite**:
- **JCI** (Job Creation Index): AI-attributable *gross job creation* (with variants).
- **JDI** (Job Displacement Index): AI-attributable *gross job destruction / displacement* (with variants).
- **NEIS** (Net Employment Impact Score): normalised net effect with uncertainty bounds.
- **AIRR** (AI Responsibility Ratio): a governance-linked “responsibility” metric with multiple credible formulations.
- A **Job Displacement Charter (JDC)**: a governance contract specifying **minimum duties** for disclosure, worker information, transition supports, monitoring, and red-teaming - aligned to internationally recognised AI and labour principles and to due-diligence expectations for responsible business conduct.

- A set of **reporting templates**, **audit checklists**, and **anti-gaming diagnostics** suitable for regulators, boards, unions, and investors.

Key institutional design principles. 1. **Use gross flows, not only net flows.** Job creation/destruction should be measured as gross components (openings/expansions vs contractions/closures), consistent with international practice (e.g., Business Employment Dynamics in the US; payroll-record methods in Ireland). 2. **Separate measurement from attribution.** JCI/JDI require a transparent distinction between:

- *Observed* job flows (administrative reality), and
- *AI attribution* (counterfactual claim), which is the main source of uncertainty and manipulation risk.

3. **Treat "job displacement" as a worker outcome as well as a job outcome.** Worker displacement definitions - e.g., the US definition tied to plant closure/move, insufficient work, abolishment of position/shift - provide an auditable basis for displaced-worker counting and follow-up outcomes. 4. **Integrate "Decent Work" constraints.** Job quality matters: the JCI-JDC framework is explicitly compatible with the ILO's Decent Work framing and its centrality of employment, rights at work, social protection, and social dialogue. 5. **Make the system audit-resistant.** The report includes "self-red-teaming" and adversarial diagnostics to detect reclassification, outsourcing arbitrage, wage-bill laundering, and timing games.

Governance pathway. Near-term adoption is feasible through (i) voluntary codes anchored in existing AI governance standards (e.g., OECD AI Recommendation), (ii) corporate reporting alignment with workforce disclosure regimes (e.g., ESRS under CSRD), and (iii) operational management systems (e.g., ISO/IEC 42001) supported by assurance and audit protocols.

Assumptions and scope constraints (explicit).

- Target jurisdictions are unspecified (assume a design intended to be interoperable across OECD/ILO member contexts and adaptable to national statistical and legal systems).
- Data access constraints vary widely (assume some deployments will rely on HR/payroll administrative data, while others rely on surveys and modelled attribution).
- The framework is designed for **advanced AI adoption**, but is technology-agnostic (AI constitutes a socio-technical system; risk management must cover lifecycle governance).

Sections A-K

Section A: Document Map and Reading Guide

Purpose and scope

This document specifies an institutional-level framework to (i) **measure** AI-related job creation and job displacement, and (ii) **govern** AI deployment so that labour-market transitions preserve **human economic agency** and meet **minimum due-diligence and reporting expectations**. The reading map uses A-K sections consistent with PAD-style charters.

The JCI-JDC is intended for:

- Governments and regulators
- International organisations (ILO/OECD/UN system)
- Corporate boards, investors, and auditors
- Labour unions and employer organisations
- Labour economists, AI governance and AI risk professionals

Core outputs and where to find them

- **Metrics + formulas (JCI/JDI/NEIS/AIRR):** Section C
- **Data collection + sampling + error model + falsifiability tests:** Sections C, D, K
- **Governance architecture + roadmap:** Sections I, K
- **Adversarial manipulation scenarios + diagnostics:** Sections G, J
- **Templates + checklists + corporate disclosure template:** Section K and Appendices

Updateability: versioning and change control

Recommendation: maintain this document as a “living standard” with:

- **Semantic versioning** (e.g., JCI-JDC 1.0, 1.1…)
- A public **change log**
- A standing **Technical Standards Committee** (statistical agencies, labour ministry, unions, employers, auditors, AI safety experts) aligned to international coordination initiatives for AI governance.

Measurement design note (boundary conditions)

Jobs and occupations should be defined consistently with international statistical concepts (e.g., ISCO’s definition of a job as a set of tasks and duties performed by one person). Coverage decisions must be explicit: employees vs contractors, domestic vs global operations, and supply-chain scope.

Falsifiability test for Section A

If readers cannot reproduce (a) metric definitions, (b) data lineage, and (c) governance responsibilities from this map without further clarification, the map is not fit-for-purpose and must be revised (test: “10-minute reconstructability” by independent reviewers).

Section B: Multi-audience Executive Summaries

B.1 Operator summary

Operators (AI product owners, HR analytics teams, data stewards, internal auditors) should treat JCI-JDC as a **control system**:

- **Instrument the organisation** so that gross job flows can be computed quarterly using payroll/HR data, analogous to official gross job gains/losses definitions used in labour market statistics.

- **Separate observed flows from AI attribution**: observed flows are administrative truth; attribution is an analytical claim with uncertainty bounds.
- Implement a minimum governance stack:
- risk management lifecycle mapping aligned to AI governance frameworks (e.g., mapping to NIST AI RMF functions and/or ISO/IEC 42001 management system controls).
- Maintain audit-ready logs for measurement, consistent with emerging regulatory expectations around logging and governance for “high-risk” AI deployments in relevant jurisdictions (as a model case).

Operator falsifiability test: A third-party auditor should be able to recompute JCI/JDI from raw extracts and obtain the same results within stated tolerances.

B.2 Institutional summary

Institutional leaders (ministries, regulators, boards, international bodies) should treat JCI-JDC as a **complement to AI governance and workforce disclosure regimes**, not as a standalone statistic:

- The OECD AI Recommendation establishes internationally recognised expectations for trustworthy AI and for preparing for labour market transformation.
- The ILO Decent Work core framing provides the normative constraints: employment creation, rights at work, social protection, and social dialogue.
- Workforce reporting is becoming structurally integrated into corporate disclosure regimes in multiple jurisdictions (e.g., ESRS under delegated EU regulations).
- Institutions should require:
- standardised reporting templates,
- independent assurance,
- anti-gaming diagnostics,
- enforceable incentives (procurement access, tax benefits, licensing conditions).

Institutional falsifiability test: A regulator should be able to detect systematic under-reporting or outsourcing arbitrage by cross-referencing payroll and procurement trends with published JCI/JDI.

B.3 Adversarial perspective summary

A motivated actor may try to “win” JCI-JDC metrics while increasing displacement harms. Likely adversarial directions:

- Reclassify employees as contractors, relocate displacement to suppliers, or delay layoffs via attrition while claiming “no job losses.”
- Overstate “AI-created” roles by relabeling existing jobs, or count temporary project roles as permanent creation.
- Inflate retraining spend without outcome accountability (“training theatre”).

This document explicitly anticipates these manipulation channels and specifies detection diagnostics and audit tests (Sections G and J), drawing on the broader logic of due diligence expectations in responsible business conduct (identify, prevent, mitigate, and account for adverse impacts).

Adversarial falsifiability test: If an operator can materially improve reported JCI/JDI without corresponding changes in payroll totals, benefits coverage, and worker transition outcomes, the system has exploitable loopholes.

B.4 Post-AGI context summary

If AI capabilities approach transformative levels, JCI-JDC becomes part of a **minimum viable governance** stack for preserving economic agency:

- Task-based economic frameworks imply that labour outcomes depend on whether AI is deployed to replace tasks vs create new tasks and complementary innovation.
- Occupational exposure indices for generative AI suggest the dominant near-term mechanism may be **task transformation/augmentation** rather than full occupation automation, but this varies by occupation and country.

Post-AGI falsifiability test: If displacement accelerates while reported JCI remains high, JCI's attribution and quality-adjustment rules must be strengthened (or the metric should be treated as invalid).

Section C: Formal Definitions

This section defines the metric suite and the charter object. Formal definitions should be consistent, auditable, and aligned with statistical standards where available. Relevant baseline statistical notions: job as tasks/duties performed by one person (ISCO framing).

C.1 Core constructs

Organisation (reporting entity): A bounded unit required to publish JCI-JDC reporting. Boundary must be explicit (legal entity vs group consolidation; domestic vs global).

Job (operational definition): A position held by an individual worker during a reference period, mapped to an occupational code and employment relationship status. Consistent referencing to ISCO concepts is recommended where cross-national comparability is required.

Employment unit: Choose one (and disclose):

- **Headcount** (persons employed)
- **FTE** (full-time equivalent)
- **Hours worked** as labour input proxy, noting that "hours actually worked" is conceptually preferred for labour input measurement in productivity contexts.

Job creation (gross): At the unit-of-observation level (establishment / enterprise / cost centre), job creation in period *t* equals positive employment change; otherwise 0 - consistent with administrative payroll-record definitions used in practice.

Job destruction (gross): The negative employment change (where employment falls), else 0, consistent with gross job losses definitions in official job-flow statistics.

Worker displacement: A worker-level outcome involving involuntary job loss/exit due to structural reasons (closure/move, insufficient work, abolished position/shift), using the displaced worker survey definition as an auditable benchmark for worker-based displacement counting.

Job Displacement Charter (JDC): A written set of commitments and constraints on AI deployment designed to:

- mandate consistent measurement and reporting of displacement/creation,
- require worker information and consultation where applicable,
- require transition supports and outcomes,
- require anti-deception controls (auditability and red-teaming).

JDC draws normative grounding from:

- international AI stewardship principles (OECD AI Recommendation),
- ILO Decent Work agenda pillars,
- and responsible business conduct due diligence expectations.

C.2 Formal metrics and formulas

Baseline gross job flows (observed)

Let *i* index reporting units (establishments, subsidiaries, cost centres), and *t* index time (quarter or year). Let $E_{i,t}$ be employment (FTE or headcount) in unit *i* at period *t*.

$$ JC_{i,t}=\max(E_{i,t}-E_{i,t-1},\,0) $$ $$ JD_{i,t}=\max(E_{i,t-1}-E_{i,t},\,0) $$

These formulas reflect standard gross flow logic used in official practice.

Aggregate observed gross flows: $$ JC_t=\sum_i JC_{i,t},\quad JD_t=\sum_i JD_{i,t} $$ Net employment change: $$ \Delta E_t = JC_t - JD_t $$

AI attribution layer (the “index” component)

Define attribution shares $a_{i,t}$ and $b_{i,t}$ in $[0,1]$, interpreted as the proportion of job creation/destruction in unit *i* attributable to AI adoption (vs other drivers).

AI-attributable job creation:

$$AJC_{i,t}=a_{i,t}\cdot JC_{i,t}$$

AI-attributable job destruction:

$$AJD_{i,t}=b_{i,t}\cdot JD_{i,t}$$

Job Creation Index (JCI) (level form):

$$JCI_t=\sum_i AJC_{i,t}$$

Job Displacement Index (JDI) (level form):

$$JDI_t=\sum_i AJD_{i,t}$$

Normalised index variants (recommended for comparability):

- Per 1,000 baseline employees:

$$JCI^{(1000)}_t = 1000\cdot \frac{JCI_t}{E_{t-1}},\quad JDI^{(1000)}_t = 1000\cdot \frac{JDI_t}{E_{t-1}}$$

- Per revenue or value added (optional; requires national-accounts-consistent labour cost concepts if used for wage-bill normalisation).

Net Employment Impact Score (NEIS)

A practical, internationally comparable score is the net effect per 1,000 baseline employment:

$$NEIS_t = JCI^{(1000)}_t - JDI^{(1000)}_t$$

Interpretation: net AI-attributable jobs created (or displaced) per 1,000 baseline jobs, subject to uncertainty bounds.

AI Responsibility Ratio (AIRR)

AIRR should be defined in at least **two complementary forms**:

AIRR-Investment (resource-based): Let TI_t be transition investment (training, redeployment, wage insurance, placement services, certified programmes), and AIU_t be AI-driven economic uplift (cost savings or incremental EBITDA attributable to AI projects under the same attribution discipline).

$$AIRR^{(inv)}_t = \frac{TI_t}{AIU_t}$$

AIRR-Outcome (worker-based): Let DW_t be displaced workers attributable to AI (worker-level), and RW_t be the number re-employed/redeployed into equal-or-better quality work within a defined horizon (e.g., 12 months), with job-quality rules defined below.

$$AIRR^{(out)}_t = \frac{RW_t}{DW_t}$$

Worker displacement measurement should be consistent with official displaced-worker counting logic where possible.

Job-quality adjustment (optional but recommended)

To align with “minimum viable human economy,” JCI-JDC should include an optional **quality-adjusted** layer:

Let q_{j} be a job-quality weight for job *j* (e.g., 0-1.5), derived from job attributes:

- contract security (permanent vs temporary),
- wage percentile or living-wage threshold,
- benefits and social protection coverage,
- working-time stability,
- occupational safety indicators.

These attributes align with the “decent work” pillar logic.

Then: $$JCI^{(Q)}_t=\sum_{j \in \text{created jobs}} q_j,\quad JDI^{(Q)}_t=\sum_{j \in \text{displaced jobs}} q_j$$ $$NEIS^{(Q)}_t=1000\cdot \frac{JCI^{(Q)}_t - JDI^{(Q)}_t}{E_{t-1}}$$

C.3 Data collection methodologies

Preferred data hierarchy (accuracy-first): 1. **Administrative payroll/HR systems** (highest fidelity for headcount/FTE, wages, tenure). 2. **Statistical business registers** as sampling frames for establishment surveys where required. 3. **Establishment surveys** for job flows, vacancies, labour turnover (where administrative capture is incomplete). 4. **Household labour force surveys** for worker outcomes and underutilisation context, aligned with international labour statistics standards. 5. **Enterprise surveys** for cross-country comparability and private-sector snapshots (World Bank Enterprise Surveys).

Attribution evidence sources (to estimate $a_{i,t}$, $b_{i,t}$):

- AI project register: deployments, business processes affected, go-live date.
- Task mapping: tasks automated vs augmented (task-based approach aligns with empirical literature and occupational exposure indices).
- Operational metrics: throughput, quality, error rates, cycle time (pre/post).
- Organisational actions: hiring freezes, backfill policies, redeployments.

C.4 Sampling strategy (when full measurement is infeasible)

Corporate level (multi-site firms):

- Stratify establishments by:
- AI intensity (high/medium/low; derived from AI project register),
- size class,
- occupation mix.
- Oversample high-AI-intensity units to bound tail risks.
- Require minimum sample sizes to support 95% confidence intervals for key rates.

National level (regulators/statistical agencies):

- Combine administrative sources and surveys; publish methodology notes on sampling and measurement error sources (sampling and measurement errors are fundamental limitations of survey-based work statistics).

C.5 Measurement error estimates and disclosure requirements

JCI-JDC must require reporting of:

- **Sampling error** (if survey-based): standard errors / confidence intervals.
- **Measurement error**: misclassification (employee vs contractor), occupational coding errors, missing locations, data lags.
- **Attribution error**: the largest component; disclose attribution method tier and sensitivity ranges.

A practical disclosure format:

- $JCI_t = \hat{JCI}_t \pm \epsilon^{(meas)} \pm \epsilon^{(attr)}$
- With $\epsilon^{(attr)}$ derived from sensitivity analysis (e.g., vary attribution shares ±20% around central estimates, unless a causal design supports tighter bounds).

C.6 Falsifiability tests (core metric falsification)

1. **Recomputation test:** Independent auditor recomputes JC/JD from payroll extracts and matches within tolerance. 2. **Cross-source reconciliation:** JCI/JDI trends must reconcile with wage-bill, benefits coverage, and procurement shifts (outsourcing). 3. **Pre-registration test:** Attribution method must be documented *before* observing outcomes (reduces post-hoc rationalisation). 4. **External plausibility test:** Unit-level JCI/JDI outliers must be explained with evidence trail (AI projects, market shocks, restructuring).

Section D: Uncertainty Taxonomy

JCI-JDC involves uncertainties in both labour measurement and AI impact attribution. International labour statistics guidance emphasises that data sources differ in strengths and limitations and that sampling and measurement errors affect quality.

D.1 Classes of uncertainty

1. **Statistical uncertainty (quantifiable):**
   - sampling variance (survey-based),
   - linkage errors in administrative longitudinal data,
   - occupational misclassification.
2. **Model uncertainty (partially quantifiable):**
   - conversion between headcount, FTE, and hours worked,
   - mapping tasks to occupations.
3. **Counterfactual/attribution uncertainty (often dominant):**

- what would have happened without AI?
- confounding shocks (demand shifts, macro cycles, restructuring).

4. **Strategic uncertainty (adversarial):**
   - deliberate manipulation of boundaries, timing, classification.
5. **Deep uncertainty (structural):**
   - general equilibrium effects (macro job creation elsewhere),
   - long-lag effects (education pipelines, entry-level erosion).

D.2 Epistemic integrity rules (institutional)

Because AI can change workplaces and information environments, institutions should adopt strict epistemic rules:

- **document evidence trails**,
- **disclose uncertainties**,
- **require independent assurance**.

This aligns with governance expectations in sustainability reporting regimes and the logic of internationally recognised AI governance initiatives.

D.3 Measurement error budget template (recommended)

Require an “error budget” table:

Error component	Symbol	Typical driver	Estimation method	Disclosure
Sampling error	ϵ_s	survey sampling	SE/CI	Mandatory if survey-based
Measurement error	ϵ_m	coding, missingness	validation sample	Mandatory
Linkage error	ϵ_ℓ	establishment ID linking	linkage QA	Mandatory for admin longitudinal
Attribution error	ϵ_a	counterfactual	sensitivity / causal design	Mandatory
Strategic manipulation risk	ϵ_{adv}	gaming	red-team tests	Mandatory

D.4 Falsifiability test for Section D

If a reporting entity cannot produce an explicit attribution uncertainty bound (even a wide one), then NEIS/AIRR cannot be treated as decision-grade and must be labelled “exploratory.”

Section E: Institutional Failure Modes

Institutional failures in labour-impact governance predictably arise from incentive conflicts, weak auditability, and coordination gaps between developers, employers, regulators, and worker representatives. Analogous failures are well documented across high-stakes domains; JCI-JDC focuses on the AI-labour domain specifically.

E.1 Principal failure modes (JCI-JDC-specific)

1. **Metric capture:** JCI becomes a “PR KPI” rather than a constraint (optimising reported job creation while offloading displacement). 2. **Boundary arbitrage:** shifting jobs to contractors/suppliers to suppress measured displacement. 3. **Attrition laundering:** eliminating roles via non-backfill policies while claiming “no layoffs.” 4. **Transition theatre:** training spend disconnected from outcomes; AIRR(inv) improves without AIRR(out) improving. 5. **Delayed recognition:** displacement costs occur with lag; short reporting windows hide impacts. 6. **Audit opacity:** lack of data lineage prevents assurance. 7. **Weak social dialogue:** worker input is absent despite its centrality to decent work and transition governance.

E.2 Diagnostics (institutional)

A robust institution should be able to answer:

- Who owns the AI deployment register and who owns employment measurement?
- Is there an independent assurance function?
- Are workers informed and consulted where required/appropriate (e.g., as modelled by legal requirements in some jurisdictions for workplace high-risk AI systems)?

E.3 Measurement and sampling implications

Institutional failure risk increases when:

- measurement relies on self-reported surveys only,
- occupational coding is inconsistent,
- attribution is not pre-registered.

E.4 Falsifiability test for Section E

If an entity repeatedly reports high JCI with flat/declining total payroll and rising vendor labour spend in affected functions, it triggers a presumption of boundary arbitrage requiring enhanced audit and restatement.

Section F: Employment Alignment Principles

Employment alignment means AI deployment should preserve a viable human economy and social stability while enabling productivity gains. This section anchors principles in internationally recognised labour and AI governance norms.

F.1 Core employment alignment principles

1. **Decent Work compatibility.** AI-related restructuring should be evaluated against the ILO's Decent Work pillars and the role of social dialogue and rights at work. 2. **Worker agency and information.** Workers should be informed when AI systems strongly shape employment decisions or workplace monitoring, and meaningful oversight should exist (a regulatory model is visible in recent AI governance regimes). 3. **Transparency and auditability.** Employment metrics must be verifiable and comparable, consistent with the direction of corporate sustainability reporting standards ecosystems. 4. **Attribution integrity.** AI attribution must be conservative, evidence-based, and sensitive to confounding factors. 5. **Transition responsibility.** Entities benefiting from automation should carry measurable responsibility for transition supports (AIRR), aligned with responsible business conduct due diligence logic. 6. **Non-displacement by stealth.** "No layoffs" is not a sufficient claim; attrition and outsourcing effects must be included in JDI.

F.2 Metric implications

- Require both **AIRR(inv)** and **AIRR(out)** to reduce "spend-only" manipulation.
- Require **quality-adjusted** options (NEIS(Q)) when workforce welfare is material.

F.3 Measurement error expectations

Employment alignment reporting should require:

- transparent job-quality weighting rules,
- disclosure of the sensitivity of job-quality weights to wage inflation, hours changes, and benefit changes.

F.4 Falsifiability test for Section F

If job-quality-adjusted indices show deterioration while unadjusted indices remain positive, employment alignment is failing in substance (even if net headcount is stable).

Section G: Threat Models

Threat models in JCI-JDC focus on (i) how AI changes labour demand, and (ii) how institutions can manipulate or misread signals.

G.1 Labour-market threat models (substantive, non-adversarial)

1. **Task displacement dominance:** AI primarily replaces labour tasks without sufficient new-task creation (displacement effect outweighs reinstatement). 2. **Entry-level pipeline collapse:** fewer junior roles reduce skill formation and social mobility (requires long-horizon indicators). 3. **Job-quality degradation:** wage stagnation, intensified work, monitoring/privacy risks - highlighted in workplace AI impact analyses. 4. **Geographic and sectoral divergence:** impacts differ by country and occupational composition; task exposure measures show heterogeneity.

G.2 Adversarial manipulation threat models (metric gaming)

Adversarial scenarios and detection diagnostics (required table).

Manipulation pattern	What the actor does	Metric impact	Detection diagnostics	Required countermeasure
Contractor relabeling	Convert employees -> contractors	Lowers JDI; raises “flexible” JCI	Rising vendor labour spend + falling payroll headcount; benefits coverage drop	Mandatory “employment boundary reconciliation” (payroll + procurement)
Attrition laundering	Hiring freeze + non-backfill	JDI understated	Vacancy fill-rate drop; workload proxy rises; output stable	Require “non-backfill displacement” reporting + workforce survey
Reclassification of existing roles	Rename jobs as “AI roles”	Inflates JCI	Job-code lineage shows no task change	Require role-change evidence and task mapping
Offshoring displacement	Move roles abroad	Domestic JDI understated	Location-coded payroll shift	Mandatory geographic scope disclosure and global reporting option
Temporary project inflation	Count short contracts as creation	JCI spikes	contract duration distribution shifts	Require duration-weighted JCI and 12-month retention rate
“Training theatre”	Spend on training without redeployment	AIRR(inv) rises, AIRR(out) flat	outcome tracking shows low placement	Require AIRR(out) and independent evaluation
“AI uplift overclaim”	Overstate AI-driven profits	AIRR(inv) looks low	finance reconciliation fails	Assurance of AIU attribution (audit rules)

Due diligence expectations and “accounting for impacts” logic support requiring such structured controls.

G.3 Measurement and sampling implications

- Oversample “high-risk functions” (HR, customer support, software dev, clerical) for displacement monitoring, consistent with occupational exposure approaches used in global analyses.
- Require “supplier employment risk” indicators where outsourcing is material (Scope 2/3 analog).

G.4 Falsifiability test for Section G

If adversarial diagnostics identify boundary arbitrage and the reporting entity cannot reconcile discrepancies within one reporting period, published JCI/JDI must be restated and assurance opinion qualified.

Section H: Minimum Viable Human Economy

This section defines what “minimum viability” means economically, not biologically: the minimum structure of employment, income distribution, and labour-market institutions required to preserve social cohesion and agency under AI acceleration.

H.1 Definition: Minimum Viable Human Economy (MVHE)

MVHE is the minimum set of conditions under which a society can maintain:

- broad-based access to income (through work, transfers, or hybrid),
- skill formation pipelines,
- social protection and labour rights enforcement,
- stable demand sufficient to avoid persistent macro instability.

This parallels the ILO’s emphasis on employment, social protection, rights at work and social dialogue as central pillars of functioning societies.

H.2 MVHE indicators (proposed)

Governments and international organisations should track:

- Employment and underutilisation measures (consistent with ILO standards).
- Displaced-worker incidence and re-employment outcomes where measured.
- Job flows (gross creation/destruction) to detect churn even in stable net employment.
- Labour share / wage bill measures (national accounts-compatible remuneration concepts).

H.3 MVHE policy constraints (JDC minimums)

A credible JDC should specify:

- minimum transition supports per displaced worker (AIRR constraints),
- requirements for social dialogue/consultation,
- priority protections for critical public services and essential skill pipelines.

H.4 Measurement error and falsifiability for MVHE

Error: MVHE indicators combine data sources with differing lags and conceptual bases; publish reconciliation notes. **Falsifiability test:** If a jurisdiction exhibits rising displacement and falling re-employment outcomes while JCI is positive, the framework must flag an MVHE risk breach and trigger policy review.

Section I: Governance Architecture

Governance must align incentives across firms, workers, regulators, auditors, and international coordination bodies. This section proposes a multi-layer architecture compatible with existing standards.

I.1 Architecture overview

- **Enterprise layer:** Board accountability + ISO/IEC 42001-style AI management system controls for AI governance maturity.
- **Labour layer:** Institutionalised social dialogue and worker participation mechanisms (Decent Work pillar logic).
- **Regulatory/statistical layer:** Harmonised definitions aligned to AI and sustainability reporting regimes (e.g., ESRS under delegated regulations), with assurance requirements.
- **International layer:** Coordination through OECD AI principles, ILO frameworks, and UN governance initiatives.

I.2 Proposed institutional roles

- **National Statistical Authority:** defines measurement standards; integrates JCI/JDI into labour-market statistical programmes where feasible.
- **Labour Regulator / Ministry:** enforces charter compliance in high-risk sectors; manages worker transition funds.
- **AI Regulator / Digital Authority:** enforces AI risk obligations; links AI deployment licensing to employment-impact reporting (where relevant).
- **Independent Assurance Providers:** audit JCI/JDI and AIRR disclosures using the audit checklists in Section K.
- **Tripartite bodies (where present):** unions/employers/government co-define job-quality weights and transition thresholds.

I.3 Mermaid diagram: Governance architecture (conceptual)

```
flowchart TB
 subgraph Enterprise
 BOD[Board / Audit Committee]
 AIMS[AI Management System\n(ISO/IEC 42001-aligned)]
 HR[HR + Workforce Analytics]
 OPS[Business Units / Operations]
 UNION[Worker Reps / Unions]
 end

 subgraph National
 STAT[National Statistical Authority]
 LAB[Labour Ministry / Regulator]
 AIReg[AI / Digital Regulator]
 AUD[Independent Assurance]
 end

 subgraph International
 OECD[OECD AI & RBC Norms]
 ILO[ILO Decent Work / Labour Stats]
 UN[UN AI Governance Initiatives]
 end

 OPS --> HR
 AIMS --> HR
 HR --> BOD
```

```
UNION --> BOD
HR --> AUD
AUD --> LAB
AUD --> AIReg
AUD --> STAT
STAT --> LAB
OECD --> STAT
ILO --> STAT
UN --> AIReg
LAB --> Enterprise
AIReg --> Enterprise
```

I.4 International governance roadmap (phased)

```
gantt
 title JCI-JDC International Roadmap (illustrative)
 dateFormat YYYY-MM
 section Phase 1: Voluntary Standardisation
 Draft metric standard + pilots :a1, 2026-04, 12m
 Assurance guidance + audit checklists :a2, 2026-06, 12m
 section Phase 2: Regulatory Integration
 National disclosure requirements (high-risk sectors) :b1, 2027-01, 18m
 Statistical harmonisation + business register alignment :b2, 2027-01, 24m
 section Phase 3: International Interoperability
 OECD/ILO alignment + model law guidance :c1, 2028-01, 24m
 Multilateral reporting interoperability (ESG + AI) :c2, 2028-06, 24m
```

This sequencing is consistent with the reality that AI governance is already moving through intergovernmental standards (OECD) and formal legal regimes (e.g., EU AI Act), while corporate reporting regimes standardise disclosure (ESRS).

I.5 Enforcement mechanisms and incentives

- **Assurance requirement:** qualified audit opinions if reconciliation fails.
- **Procurement incentives:** public procurement eligibility conditional on disclosure (common good logic).
- **Tax incentives:** conditional AI tax credits tied to AIRR thresholds.
- **Penalties for deception:** civil penalties for material misstatement in workforce impact reporting (analogous to sustainability disclosure enforcement logic).

I.6 Falsifiability test for Section I

If governance cannot prevent boundary arbitrage (persistent divergence between payroll-based and vendor-based labour costs) the architecture is insufficient; enforcement and scope rules must be tightened.

Section J: Self-Red-Teaming

This section identifies weaknesses, loopholes, and failure points in JCI-JDC itself, and prescribes robustness tests.

J.1 Ambiguities and loopholes

- **Attribution ambiguity:** Without strict attribution tiers, JCI can drift into narrative.
- **Quality-weight controversy:** Job-quality weights are normative; they must be documented and co-governed (e.g., tripartite).
- **Global vs domestic scope:** Domestic-only reporting can hide offshoring displacement.
- **Supplier scope:** Without supply-chain scope, outsourcing arbitrage survives.

J.2 Adversarial exploitation tests (required)

- “Shadow workforce” test: Can the firm materially reduce payroll headcount while increasing contractor labour without triggering JDI?
- “Role relabel” test: Can the firm reclassify roles to claim JCI without task evidence?
- “Training theatre” test: Can AIRR(inv) rise while AIRR(out) stays flat?

J.3 Assumptions and falsifiability conditions

Assumption: Administrative data is sufficiently high quality for job flows. Falsification: payroll extracts cannot be linked longitudinally or have missing key fields. (Mitigation: require minimum data schema; use establishment survey backup.)

Assumption: AI attribution can be bounded. Falsification: attribution swings wildly under reasonable sensitivity ranges; treat NEIS as non-decision-grade in that case.

Assumption: Social dialogue improves transition outcomes. Falsification: no measurable improvement under strong participation; revise programme design (but maintain rights-based rationale).

J.4 Robustness under extreme scenarios

- **Rapid adoption shock:** require quarterly reporting, not annual.
- **Sectoral collapse:** activate labour-market adjustment funds based on JDI thresholds.
- **Regulatory arbitrage:** require cross-border reporting standardisation via international alignment.

Section K: Implementation Protocols

This section provides practical protocols, templates, checklists, and corporate disclosure requirements.

K.1 Minimum implementation steps (enterprise)

1. **Governance setup**
    - Assign a Board sponsor and a responsible executive.

- Establish worker consultation mechanisms consistent with local law and good practice; where workplace AI is “high-risk,” ensure worker information requirements are operationalised (model case: EU high-risk workplace AI information duties).

2. **Data mapping and boundary specification**
   - Define reporting boundary, workforce taxonomy, vendor labour boundary.
   - Establish job/occupation coding rules (ISCO crosswalk where required).
3. **Baseline measurement (t0)**
   - Compute observed JC_t, JD_t from HR/payroll data.
   - Produce workforce composition tables (occupation, contract type, location).
4. **Attribution tier selection and pre-registration**
   - Tier 1: documentary attribution (AI project register + management evidence).
   - Tier 2: quasi-experimental (matched units; diff-in-diff).
   - Tier 3: task-based causal modelling (advanced).
5. **Quarterly reporting**
   - Publish JCI/JDI/NEIS + error budget + AIRR.
   - Publish reconciliation statements and material changes.
6. **Independent assurance**
   - Annual assurance opinion with limited assurance quarterly.

K.2 Sampling protocols (when needed)

- Minimum sample sizes by stratum; oversample high-AI and high-displacement-risk functions.
- Use business registers as frames where official systems exist.

K.3 Reporting templates (institutional)

Template: JCI-JDC Quarterly Disclosure (1-2 pages) 1) Reporting boundaries and workforce coverage 2) Observed gross job flows (JC/JD) 3) AI attribution method (tier + evidence) 4) JCI/JDI/NEIS with uncertainty bounds 5) AIRR(inv) and AIRR(out) 6) Job-quality adjustments (if used) 7) Worker transition outcomes (placements, wage changes, retention) 8) Red-team findings and anti-gaming diagnostics 9) Independent assurance statement status

K.4 Audit checklist (board / assurance)

Metric integrity

- [ ] Payroll data lineage documented (system-of-record, extraction procedure)
- [ ] Longitudinal linkage validated (unit IDs stable)

- [ ] JC/JD recomputation matches within tolerance

Boundary integrity

- [ ] Contractor/outsourcing labour reconciliation performed (vendor spend + contractor FTE estimate)
- [ ] Geographic scope declared; offshoring shifts disclosed

Attribution integrity

- [ ] Attribution tier disclosed and pre-registered
- [ ] Sensitivity analysis published
- [ ] Confounders explicitly discussed (demand shocks, restructuring)

Outcome integrity

- [ ] AIRR(out) computed on displaced worker cohort with defined horizon
- [ ] Job-quality rubric documented and applied consistently
- [ ] Independent evaluation of training effectiveness (not just spend)

Compliance interoperability

- [ ] For entities under sustainability reporting regimes, ensure workforce disclosures are consistent with applicable standards (e.g., ESRS structures where applicable).
- [ ] AI governance system aligned to recognised management and risk frameworks (ISO/IEC 42001, NIST AI RMF).

K.5 Corporate disclosure template (public-facing)

JCI-JDC Corporate Statement (Annual) A. Strategy and governance

- Board oversight model; AI governance management system alignment (e.g., ISO/IEC 42001)
- Scope of AI deployments affecting work, by function and geography

B. Metrics

- Observed gross job creation/destruction (JC/JD)
- JCI/JDI/NEIS with error bounds
- AIRR(inv) and AIRR(out)
- Job-quality adjusted metrics (optional)

C. Worker transition commitments

- Minimum transition package per displaced worker (training + income support + placement)
- Social dialogue/consultation approach (structures, frequency)

D. Adversarial risk and controls

- Boundary arbitrage controls and assurance coverage
- Material incidents of mismeasurement or restatements

E. Assurance

- Named assurer; opinion type; scope limits

K.6 Minimum viable regulatory instruments (menu)

- Mandatory disclosure for “high workforce-impact AI deployers” (thresholds by size, AI spend, or sector).
- Licensing/registration linkage for high-risk workplace AI impacts (jurisdiction-dependent; EU AI Act provides a template of operator obligations).
- Anti-misstatement penalties and public restatement requirements.
- Incentives: credits and procurement preferences tied to AIRR(out).

K.7 Falsifiability test for Section K

If independent assurance cannot reproduce reported JCI/JDI from raw sources, the entity must suspend “decision-grade” claims and publish a remediation plan within one reporting cycle.

Appendices

Metric variants comparison table

Metric	Variant	Best for	Main risk	Recommended default
JCI/JDI	Headcount-based	simplicity, HR reporting	ignores hours and quality	Use with disclosure of hours trends
JCI/JDI	FTE-based	part-time heavy sectors	FTE conversion inconsistencies	Preferred corporate default
JCI/JDI	Hours-based	productivity-linked analysis	hours reporting quality varies	Use when hours actually worked available
NEIS	per 1,000 baseline	comparability	sensitive to attribution	Default for cross-entity comparisons
AIRR	investment-based	budgeting and governance	training theatre	Use alongside outcome-based
AIRR	outcome-based	worker welfare	requires cohort tracking	Mandatory where displacement is material

Key reference anchor points (selected, non-exhaustive)

- **Job flows measurement (gross gains/losses):** US Business Employment Dynamics definitions and series; job creation/destruction measurement logic.
- **Worker displacement definition:** BLS displaced worker survey definition.
- **Task-based automation framework:** task displacement vs reinstatement/new tasks.
- **GenAI exposure measurement:** ILO working papers on generative AI and occupational exposure indices.
- **AI governance principles:** OECD Recommendation on AI.
- **Decent Work normative constraints:** ILO mission and Decent Work agenda pillars.
- **Corporate due diligence expectation:** OECD Guidelines for Multinational Enterprises on Responsible Business Conduct.
- **AI management system standard:** ISO/IEC 42001.
- **Corporate sustainability reporting standardisation:** EU delegated regulation for ESRS.
- **AI legal instrument example:** EU AI Act Regulation (EU) 2024/1689.

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Change log and update process

Version 1.0 (2026-03-05, Asia/Singapore reference date): initial institutional specification of JCI/JDI/NEIS/AIRR metrics, data hierarchy, anti-gaming diagnostics, and governance architecture aligned to OECD/ILO/UN/ISO/NIST reference frameworks and selected regulatory regimes as design anchors.

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